

Q1 - 25 June - Shift 1

The terminal velocity (v_t) of the spherical rain drop depends on the radius (r) of the spherical rain drop as:-

- (A) $r^{1/2}$ (B) r
(C) r^2 (D) r^3

Space for your notes:

Q2 - 25 June - Shift 1

The velocity of upper layer of water in a river is 36 kmh^{-1} . Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2} . Depth of the river is _____ m. (Co-efficiency of viscosity of water is 10^{-2} Pa.s)

Space for your notes:

Q3 - 26 June - Shift 1

An ideal fluid of density 800 kgm^{-3} , flows smoothly through a bent pipe (as shown in figure)

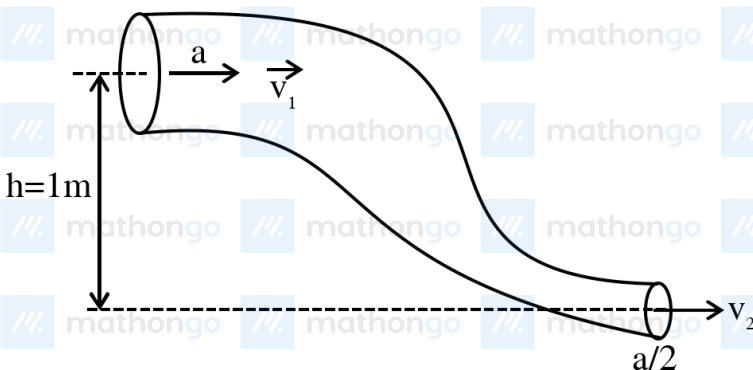
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that tapers in cross-sectional area from a to $\frac{a}{2}$. The

pressure difference between the wide and narrow sections of pipe is 4100 Pa . At wider section, the

velocity of fluid is $\frac{\sqrt{x}}{6} \text{ ms}^{-1}$ for $x = \dots\dots\dots$

(Given $g = 10 \text{ m}^{-2}$)



Q4 - 26 June - Shift 2

If ρ is the density and η is coefficient of viscosity of fluid which flows with a speed v in the pipe of diameter d , the correct formula for Reynolds number R_e is :

(A) $R_e = \frac{\eta d}{\rho v}$ (B) $R_e = \frac{\rho v}{\eta d}$

(C) $R_e = \frac{\rho v d}{\eta}$ (D) $R_e = \frac{\eta}{\rho v d}$

Space for your notes:

Q5 - 27 June - Shift 1

The velocity of a small ball of mass 'm' and density d_1 , when dropped in a container filled with glycerine, becomes constant after some time. If the density of glycerine is d_2 , then the viscous force acting on the ball, will be :

(A) $mg \left(1 - \frac{d_1}{d_2}\right)$ (B) $mg \left(1 - \frac{d_2}{d_1}\right)$

(C) $mg \left(\frac{d_1}{d_2} - 1\right)$ (D) $mg \left(\frac{d_2}{d_1} - 1\right)$

Space for your notes:

Q6 - 27 June - Shift 1

Questions

MathonGo

The area of cross-section of a large tank is 0.5 m^2 .

It has a narrow opening near the bottom having area of cross-section 1 cm^2 . A load of 25 kg is applied on the water at the top in the tank.

Neglecting the speed of water in the tank, the velocity of the water, coming out of the opening at the time when the height of water level in the tank

is 40 cm above the bottom, will be _____ cms^{-1} .

[Take $g = 10 \text{ ms}^{-2}$]

Space for your notes:

Q7 - 27 June - Shift 2

When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a

velocity v and then sinks to the bottom with the constant velocity v . It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is :

(A) 19.6 m (B) 29.4 m

(C) 39.2 m (D) 73.5 m

Space for your notes:

Q8 - 28 June - Shift 1

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Questions

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Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Product of Pressure (P) and time (t) has the same dimension as that of coefficient of viscosity.

Reason R:

Coefficient of viscosity = $\frac{\text{Force}}{\text{Velocity gradient}}$

Question: Choose the correct answer from the options given below :

- (A) Both A and R true, and R is correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Space for your notes:

Q9 - 28 June - Shift 1

A water drop of diameter cm is broken into 64 equal droplets. The surface tension of water is 0.075 N/m. In this process the gain in surface energy will be :

- (A) $2.8 \times 10^{-4} \text{ J}$
- (B) $1.5 \times 10^{-3} \text{ J}$
- (C) $1.9 \times 10^{-4} \text{ J}$
- (D) $9.4 \times 10^{-5} \text{ J}$

Space for your notes:

Q10 - 28 June - Shift 2

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Questions

MathonGo

A water drop of radius $1\mu\text{m}$ falls in a situation where the effect of buoyant force is negligible. Coefficient of viscosity of air is $1.8 \times 10^{-5} \text{ Nsm}^{-2}$ and its density is negligible as compared to that of water 10^6 gm^{-3} . Terminal velocity of the water drop is:

(Take acceleration due to gravity = 10 ms^{-2})

- (A) $145.4 \times 10^{-6} \text{ ms}^{-1}$ (B) $118.0 \times 10^{-6} \text{ ms}^{-1}$
(C) $132.6 \times 10^{-6} \text{ ms}^{-1}$ (D) $123.4 \times 10^{-6} \text{ ms}^{-1}$

Space for your notes:

Q11 - 28 June - Shift 2

A liquid of density 750 kgm^{-3} flows smoothly through a horizontal pipe that tapers in cross-sectional area from $A_1 = 1.2 \times 10^{-2} \text{ m}^2$ to

$A_2 = \frac{A_1}{2}$. The pressure difference between the

wide and narrow sections of the pipe is 4500 Pa .

The rate of flow of liquid is $\text{_____} \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$.

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Q12 - 29 June - Shift 2

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Questions

MathonGo

A small spherical ball of radius 0.1 mm and density 10^4 kg m^{-3} falls freely under gravity through a distance h before entering a tank of water. If after entering the water the velocity of ball does not change and it continues to fall with same constant velocity inside water, then the value of h will be _____ m.

(Given $g = 10 \text{ ms}^{-2}$, viscosity of water = $1.0 \times 10^{-5} \text{ N-sm}^{-2}$).

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Answer Key**Q1 (C)****Q2 (100)****Q3 (363)****Q4 (C)****Q5 (B)****Q6 (300)****Q7 (B)****Q8 (C)****Q9 (A)****Q10 (D)****Q11 (24)****Q12 (20)**

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Q1 (C)

$$v_t = \frac{2}{9} \frac{gr^2(\rho_p - \rho_l)}{\eta}; \quad v_t \propto r^2$$

Q2 (100)

$$F = \eta A \frac{\Delta v_x}{\Delta y}$$

$$\frac{F}{A} = \eta \frac{\Delta v_x}{\Delta y}$$

$$\Rightarrow 10^{-3} = 10^{-2} \times \frac{36 \times 1000}{h \times 3600}$$

$$\Rightarrow h = 10^{-2} \times \frac{36 \times 1000}{10^{-3} \times 3600} = 100 \text{ m}$$

Q3 (363)

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From continuity equation

$$av_1 = \frac{a}{2}v_2$$

$$v_2 = 2v_1$$

From Bernoulli's theorem,

$$P_1 + \rho gh_1 + \frac{1}{2}\rho v_1^2 = P_2 + \rho gh_2 + \frac{1}{2}\rho v_2^2$$

$$P_1 - P_2 = \rho \left[\left(\frac{v_2^2 - v_1^2}{2} \right) + g(h_2 - h_1) \right]$$

$$4100 = 800 \left[\left(\frac{4v_1^2 - v_1^2}{2} \right) + 10 \times (0 - 1) \right]$$

$$\frac{41}{8} + 10 = \frac{3v_1^2}{2}$$

$$\frac{121}{8} \times \frac{2}{3} = v_1^2$$

$$v_1 = \sqrt{\frac{121}{4 \times 3}} \times \frac{3}{3}$$

$$v_1 = \frac{\sqrt{363}}{6} \text{ m/s}$$

$$X = 363.$$

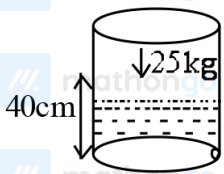
Q4 (C)

Reynold's number is given by $\frac{\rho v d}{\eta}$

Q5 (B)

$$\begin{aligned}
 F_V &= mg - F_B \\
 &= mg - \left(\frac{m}{d_1} \times d_2 \right) g \\
 &= mg \left(1 - \frac{d_2}{d_1} \right)
 \end{aligned}$$

Q6 (300)



$$P_0 + \frac{250}{0.5} + \rho g (40 \times 10^{-2}) = P_0 + \frac{1}{2} \rho v^2$$

$$500 + \frac{1000 \times 10 \times 40}{100} = \frac{1}{2} \times 1000 \times v^2$$

$$V = 3 \text{ m/s}$$

$$V = 300 \text{ cm/s}$$

Q7 (B)

$$V^2 = 2 \times 9.8 \times 4.9$$

$$V = 9.8 \text{ m/s}$$

Depth = distance travelled in 3 seconds

$$= 9.8 \times 3 = 29.4 \text{ m}$$

Q8 (C)

Pressure and time

$$P : \frac{N}{m^2}, \text{Time} : \text{Sec}$$

$$Pt = \frac{N \text{ sec}}{m^2}$$

$$\eta = \frac{F}{6\pi r v} : \frac{N}{m \cdot m / \text{sec}} : \frac{N \text{ sec}}{m^2}$$

Q9 (A)

$$d = 2\text{ cm}; \quad r = 1\text{ cm}; \quad T = 0.075$$

$$\Delta SE = T \Delta A$$

$$= 0.075 (A_f - A_i)$$

$$A_i = 4\pi r^2$$

$$A_f = 4\pi r_0^2 \times 64$$

By volume conservation

$$\frac{4}{3}\pi r^3 = 64 \cdot \frac{4}{3}\pi r_0^3$$

$$r_0 = \frac{r}{4}$$

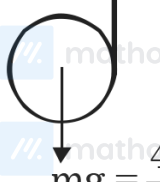
$$A_f = 4\pi \left(\frac{r}{4}\right)^2 \cdot 64 = 16\pi r^2$$

$$\Delta SE = 0.075 (16\pi r^2 - 4\pi r^2)$$

$$= 0.075 (12\pi (0.01)^2)$$

$$= 2.8 \times 10^{-4} \text{ J}$$

Q10 (D)



$$F_v = 6\pi\eta r v_t$$

$$mg = \frac{4}{3}\pi r^3 \rho g$$

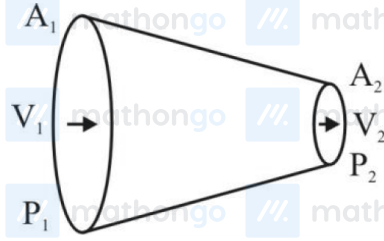
$$6\pi\eta r v_t = \frac{4}{3}\pi r^3 \rho g$$

$$v_t = \frac{4}{3} \times \frac{\pi r^3 \rho g}{6\pi\eta r}$$

$$v_t = \frac{4}{3} \times \frac{\pi r^3 \rho g}{6\pi\eta r} = \frac{2 \times 10^{-12} \times 10^3 \times 10}{9 \times 1.8 \times 10^{-5}}$$

$$= 123.4 \times 10^{-6} \text{ m/s}$$

Q11 (24)



$$A_2 = \frac{A_1}{2}$$

$$P_1 - P_2 = 4500 \text{ Pa}$$

$$P_1 + \frac{1}{2} \rho V_1^2 + \rho gh = P_2 + \frac{1}{2} \rho V_2^2 + \rho gh$$

$$P_1 - P_2 = \frac{1}{2} \rho (V_2^2 - V_1^2) \quad \dots(1)$$

$$\text{And } A_1 V_1 = A_2 V_2$$

$$\Rightarrow V_2 = 2V_1 \quad \dots(2)$$

$$4500 = \frac{1}{2} \times 750 \times 3V_1^2$$

$$V_1 = 2 \text{ m/s}$$

$$\text{Volume flow rate} = A_1 V_1 = 24 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$$

Q12 (20)

Speed after falling through height h

Should be equal to terminal velocity

$$\sqrt{2gh} = \frac{2r^2(d - \rho)g}{9\eta}$$

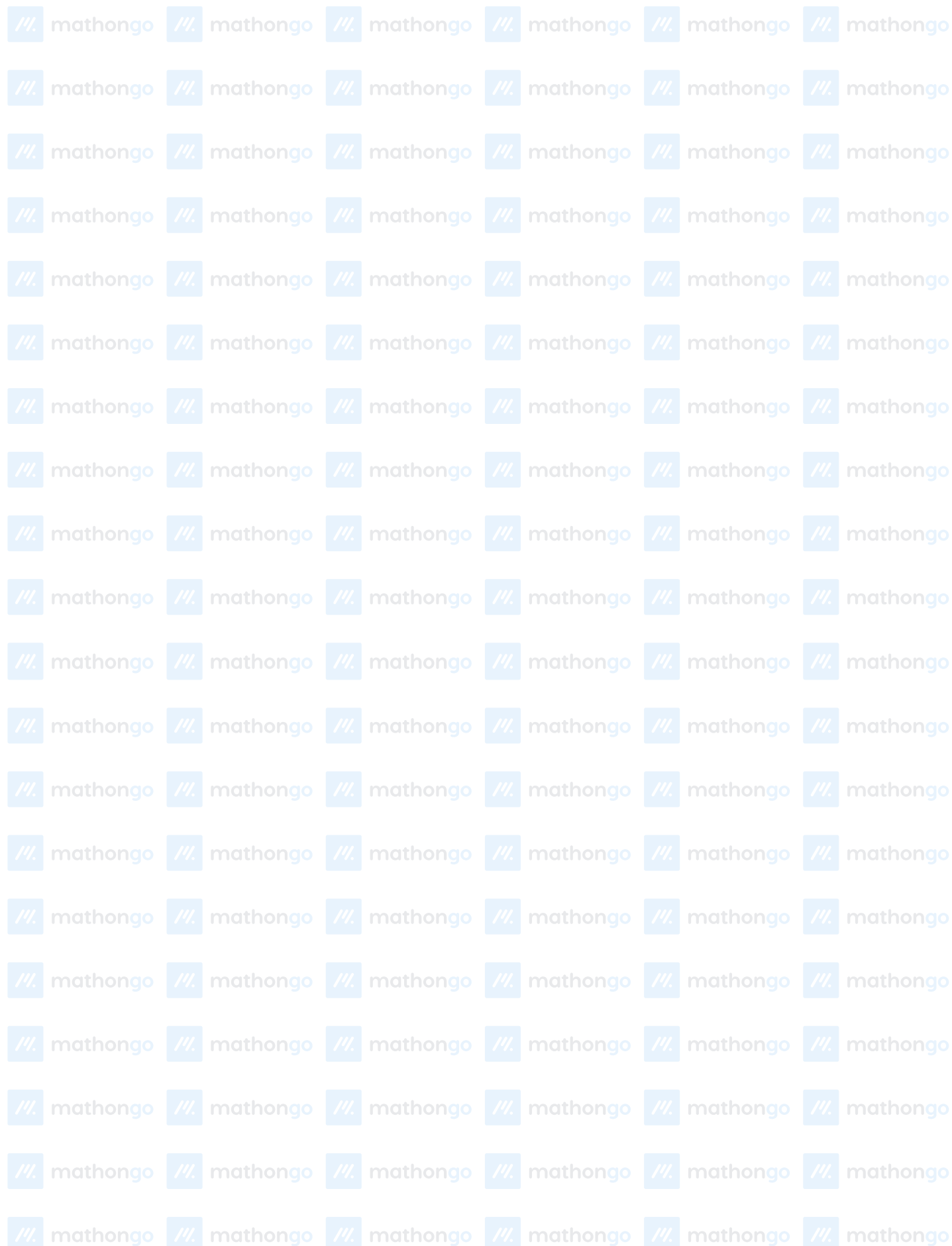
$$\sqrt{2gh} = \frac{2 \cdot 10^{-8} (10000 - 1000) \times 10}{9 \cdot 10^{-5}}$$

$$= \frac{2}{9} \times 10^{-8} \frac{9 \times 10^4}{10^{-5}} = 20$$

$$2 \times 10 \times h = 400$$

$$h = 20 \text{ m}$$

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